

Geospatial Foundation Models Benchmarking

Datasets and Evaluation Challenges

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1 Current State of GFM Benchmarking Literature

Corley et al. [1] audited 152 papers related to Geospatial Foundation Models (GFMs) and argue that the current literature is substantially less comparable and reproducible than individual papers may suggest. Their study provides a systematic snapshot of the current GFM evaluation landscape.

The corpus also highlights some ambiguity in the use of the term “foundation model”. Only 46% of the 152 papers explicitly describe their proposed model as a foundation model in the title, abstract, or contributions. The remaining papers are earlier self-supervised remote-sensing models that previous GFM surveys already consider part of the same research lineage. The authors therefore treat the 46% figure as descriptive rather than as a criterion for inclusion.

The corpus was seeded from previous GFM surveys and expanded through two additional searches: an OpenAlex and Semantic Scholar citation-graph expansion, aimed mainly at recovering 2024–2025 papers not covered by the surveys, and a keyword search over remote-sensing self-supervised learning papers published between 2019 and 2023, before “foundation model” became common terminology in the field.

The resulting corpus contains 152 papers published between 2019 and 2025. Papers from 2026 were excluded because the year was incomplete at the time of submission. Metadata extraction was LLM-assisted and subsequently manually validated by the authors [1].

1.1 Model Weights Are Often Unavailable

Across the corpus, 39% of papers release no model weights [1]. A further 19% provide a public code repository but no released model artifact, meaning that reuse would require attempting to reproduce the original pretraining procedure.

This represents a basic obstacle to reproducible benchmarking. Before a model can be evaluated on a common downstream benchmark or used as a rerun baseline, the corresponding pretrained checkpoint must first be available.

1.2 The Field Does Not Converge on a Shared Core of Benchmarks

The 152 papers report evaluations on 401 distinct benchmarks, corresponding to 1,046 evaluation experiments in total [1].

The three most frequently used benchmarks—EuroSAT [2], NWPU-RESISC45 [3], and AID [4]—account for only 10.6% of all evaluations. The remaining 89.4% is distributed across 398 benchmarks, most of which appear in only one or two papers.

The problem is not simply that some benchmarks are more popular than others. Rather, individual papers frequently evaluate on almost disjoint benchmark sets.

For each paper, Corley et al. [1] compute the fraction of its downstream benchmarks that overlap with the ten most commonly used benchmarks in the corpus. The mean overlap is only 0.27, and 50 out of 143 papers, approximately 35%, have no overlap at all with the top ten.

At the corpus level, benchmark usage has a Gini coefficient of approximately 0.51. Moreover, this coefficient remains approximately stable after 2023, while the number of benchmarks appearing in only one paper increased from 13 in 2022 to 98 in 2025. The evidence therefore does not suggest convergence towards a shared evaluation core.

This creates a GFM-specific *benchmark lottery*: because models are rarely evaluated on sufficiently overlapping benchmark sets under common protocols, the published literature does not provide enough evidence to construct a reliable field-wide ranking of GFMs [1].

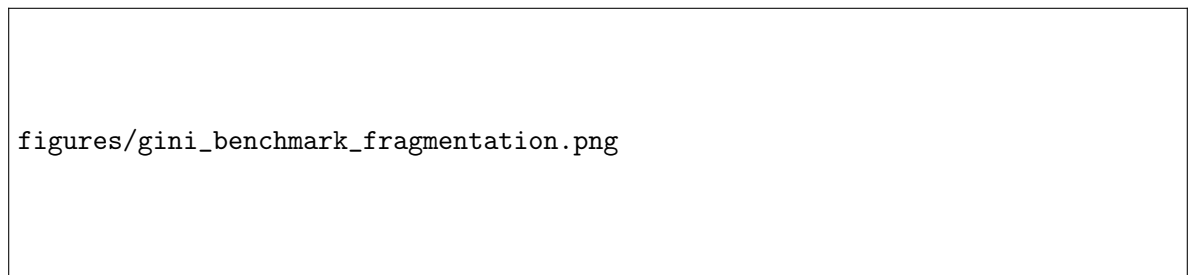


Figure 1: Benchmark usage across the audited GFM literature. The figure reports the most frequently used downstream benchmarks, per-paper overlap with the field’s top-10 benchmarks, and the evolution of benchmark concentration over time. Adapted from Corley et al. [1].

1.3 Reported Metrics Can Diverge Substantially Under the Same Nominal Protocol

The authors extracted every (model, benchmark, metric, evaluation protocol, training regime) tuple from the corpus, obtaining 10,817 results after benchmark and metric normalisation [1].

Evaluation protocols were grouped into categories such as fine-tuning, linear probing, kNN probing, zero-shot, and few-shot evaluation.

After filtering, 301 model–benchmark–protocol combinations appeared in at least two different papers. For each combination, the authors compute

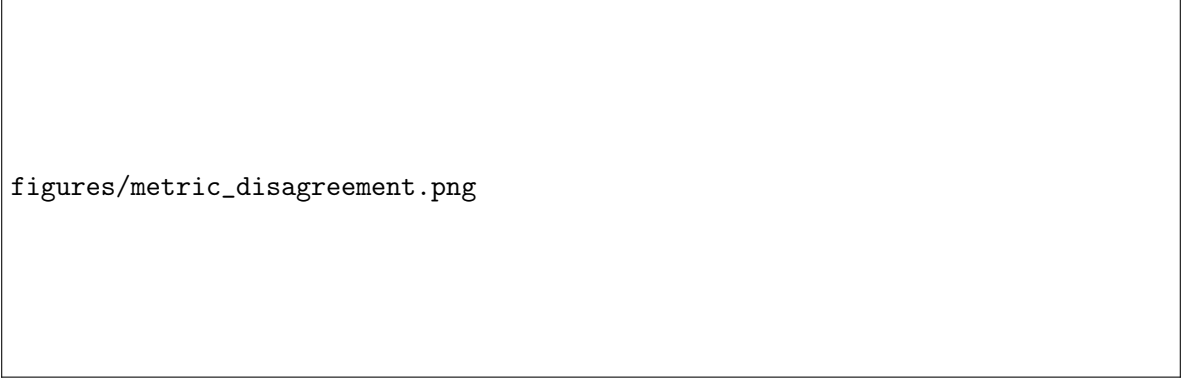
$$\text{spread} = \max(\text{reported score}) - \min(\text{reported score}).$$

Of these 301 combinations, 76 have a spread of at least 5 percentage points, 46 at least 10 points, and 20 at least 20 points [1].

The largest discrepancy concerns Scale-MAE [5] evaluated on NWPU-RESISC45 using a linear probe. One paper reports an accuracy of 33.0, whereas the original Scale-MAE paper reports 89.6, corresponding to a difference of 56.6 percentage points, despite the use of the same released ViT-L checkpoint and the same nominal evaluation protocol [1].

This example illustrates that the label “linear probe” does not uniquely specify an experiment. Hidden choices may include the optimiser, learning rate, number of epochs, image resizing or cropping, normalisation, data augmentation, representation layer, train/validation/test split, class subset, sensor channels, and other adaptation details.

A second example reported by the authors concerns GPT-4o on UCMerced under zero-shot evaluation, where reported accuracy ranges from 43.5 to 88.8 percentage points. The corresponding papers do not describe the prompting configuration in sufficient detail to explain the discrepancy [1].



figures/metric_disagreement.png

Figure 2: Largest cross-paper disagreements for repeated model–benchmark–protocol combinations. Even when the model, benchmark, and nominal evaluation protocol are the same, reported results can differ substantially. Adapted from Corley et al. [1].

Could This Be Explained by Random-Seed Variation?

The authors argue that random-seed variation alone is unlikely to explain the largest discrepancies.

Most repeated experiments agree closely, with a median spread close to zero, but the 90th-percentile spread is 12.7 percentage points [1].

Such a large tail is substantially greater than the run-to-run variation normally expected from simple classification heads under otherwise fixed conditions. The authors therefore interpret the largest disagreements primarily as evidence of hidden protocol differences.

1.4 Lack of a Shared Evaluation Harness

The geospatial community already provides benchmark collections and software infrastructure such as GEO-Bench [6], PANGAEA [7], and TorchGeo [8].

However, Corley et al. [1] argue that these resources do not yet constitute a community-wide evaluation harness comparable to those used in more mature evaluation ecosystems.

A benchmark collection may provide datasets, predefined splits, dataloaders, and example training scripts while still leaving substantial choices to the researcher. Consequently, two papers may both state that they evaluated on the same benchmark while actually executing different downstream experiments.

A stricter evaluation harness would additionally specify versioned task definitions, reference evaluation protocols, metrics, and automated checks, allowing a result to be reproduced from a model identifier and a task configuration.

Conceptually,

$$\text{model} \rightarrow \text{shared evaluation harness} \rightarrow \text{standardised result.}$$

Such a harness can reduce protocol-related degrees of freedom, but it cannot guarantee that the benchmark itself measures the intended deployment capability.

In particular, spatial dependence between training and test samples can inflate remote-sensing accuracy estimates [9, 10]. A perfectly reproducible evaluation may therefore still provide a misleading estimate of geographic generalisation.

Standardisation should also not necessarily imply forcing numerically identical hyperparameters on every architecture. Different models may require different optimisation settings. A fair comparison may instead require a common validation procedure and a comparable hyperparameter-search or compute budget, while allowing the final selected hyperparameters to differ between models.

1.5 Architecture and Pretraining Data Are Confounded

Another source of ambiguity arises when a new model simultaneously changes its architecture, pretraining data, and learning algorithm.

If several factors change at once, the observed downstream improvement cannot be attributed to one of them in isolation [1].

This is not an argument for requiring all GFMs to use identical pretraining data. Different, larger, private, or more diverse datasets may themselves be important contributions.

The problem arises when a paper attributes an improvement to an architectural or algorithmic modification without providing a controlled experiment in which the pretraining data are held fixed.

Corley et al. [11] show that apparent advantages of pretrained remote-sensing models over supervised ImageNet baselines can shrink substantially once preprocessing and distributional factors are controlled. The *No One Knows* study uses this as evidence that architecture and pretraining-data effects can be difficult to disentangle [1].

Similarly, Kaur et al. [12] study the effect of pretraining-data composition and show that changing the pretraining dataset alone can shift downstream performance by margins comparable to those often attributed to architectural novelty.

The corpus-level statistics make this comparability problem particularly clear.

Among the 126 papers for which a specific pretraining dataset could be extracted, the authors identify 87 distinct named primary pretraining datasets. Only 32 of the 126 papers, approximately 25%, share their complete pretraining configuration with at least one other paper. The remaining 94 use a configuration not shared by any other work in the corpus [1].

Consequently, comparisons in the GFM literature often involve systems that differ simultaneously in architecture, pretraining data, training objective, and downstream evaluation protocol.

Such comparisons remain useful at the *system level*, but they cannot by themselves establish which individual design choice caused a performance improvement.

1.6 Recommendations

Corley et al. [1] propose six recommendations aimed at improving comparability across the field:

1. **Release reusable model weights**, or explicitly state why they cannot be released.
2. **Evaluate on a minimum shared core of benchmark tasks**, while adding task-specific evaluations when needed.
3. **Distinguish copied and independently rerun baseline results**, including the corresponding source or configuration.
4. **Report uncertainty or repeated-run variability** when computationally feasible.
5. **Develop a shared third-party evaluation harness** with versioned protocols, common task definitions, and automated checks.
6. **Disentangle pretraining data, architecture, and learning algorithm** through controlled comparisons when causal claims are made.

2 Benchmark Datasets

2.1 Dataset Availability

Several efforts have already been made to catalogue, organise, and provide access to Earth Observation datasets for machine learning applications.

Schmitt et al. [13] provide a broad review of datasets developed for deep learning in Earth Observation, discussing their evolution across different sensors, tasks, spatial resolutions, and application domains. The same work introduces the IEEE GRSS Earth Observation Database (EOD), a community-driven catalogue intended to centralise metadata about publicly available remote-sensing datasets.

The EOD [14] provides an interactive search interface through which datasets can be filtered according to properties such as sensor modality, task, geographic location, dataset name, and keywords. This makes it a useful starting point for identifying candidate datasets satisfying specific experimental requirements.

A complementary effort is EarthNets [15], which reviews more than 500 publicly available remote-sensing datasets and organises them according to attributes including research domain, resolution, size, and relationships between datasets. Based on these attributes, the authors additionally select five large-scale datasets to form a benchmark and release the EarthNets platform for consistent model evaluation.

TorchGeo [8] provides a different but particularly useful resource from an experimental perspective. Rather than acting mainly as a dataset catalogue, TorchGeo implements a large collection of geospatial datasets through a common PyTorch interface, including benchmark datasets for classification, regression, semantic segmentation, object detection, instance segmentation, time-series analysis, and change detection. Many datasets can therefore be downloaded, loaded, and integrated into training pipelines with substantially less dataset-specific preprocessing code. This makes TorchGeo particularly relevant when selecting datasets that must later be evaluated across multiple models under a common experimental pipeline.

Additional community-maintained suites include `awesome-satellite-imagery-datasets` [16] and the `satellite-image-deep-learning/datasets` repository [17]. These resources are primarily useful for dataset discovery rather than as standardised benchmarking frameworks.

2.2 Consolidated Dataset Inventory

Table 1: Consolidated inventory of individual Earth Observation datasets identified in the surveyed sources through 2023. Task codes are used to keep the table in portrait orientation. Dataset names are linked directly when a verified project/download page was available; otherwise the superscript EOD opens the IEEE GRSS EOD search catalogue.

Task	Dataset	Year	GB	Modality / sensor	Samples	Sample size	Resolution	Classes
AE	Okutama-Action ^{EOD}	2017	25.9	RGB	77000	3840x2160	UAV@10-45m	12
AG	Paddy Rice Labeling South Korea ^{EOD}	2022	0.0016	Sentinel-2	—	—	—	—
AG	Paddy Rice Maps South Korea ^{EOD}	2022	0.198	Sentinel-1	12942	256	10m	—
AQ	Air Quality e-Reporting ^{EOD}	2021	—	—	—	—	—	—
AD	Aerial Anomaly Detection ^{EOD}	2022	0	RGB	—	—	UAV	2
BLD	Urban 3D Challenge ^{EOD}	2017	—	RGB	157000	—	0.5m	2
CD	DFC09 ^{EOD}	2009	0.098	Multiple (Satellite)	2	—	—	—
CD	Abenberg ALS ^{EOD}	2013	0.258	ALS	5400000	—	16 pts/m2	—
CD	Cross-sensor Bas-trop ^{EOD}	2015	—	MS	4 444x300,1534x300	1534x300	1534x300	2
CD	AIST Building CD	2017	1.82	Optical (Aerial)	16950	160	—	1
CD	AIST Building Change Detection ^{EOD}	2017	17.77	RGB	16950	160	0.4m	2
CD	HSI CD ^{EOD}	2017	1.7	Hyperspectral (Multiple)	3*224	800	—	5
CD	AICD ^{EOD}	2018	1.7	Optical (Aerial)	1000	800	—	—
CD	CDD (season-varying) ^{EOD}	2018	2.7	RGB	16000	256	0.03~0.1m	2
CD	GETNET dataset ^{EOD}	2018	0.05	Hyperspectral	1	463x241	30m	2
CD	Hermiston City Ore-gon ^{EOD}	2018	—	Hyperspectral	1	390x200	30m	5
CD	Hyperspectral CD ^{EOD}	2018	1.7	Hyperspectral	3 984x740,600x500	390x200	390x200	5
CD	Onera Satellite CD	2018	0.48	Sentinel-2	24	600	10m	2
CD	OneraCD	2018	0.489	Multispectral (Satellite)	24	600	—	2
CD	Synthetic & Real ^{EOD}	2018	2.7	Optical (Aerial)	16000	256	—	—
CD	WHU Building change detec-tion ^{EOD}	2018	5	RGB	2 15354x32507	0.075m	0.075m	2
CD	xView2	2018	51	RGB	22068	1024	0.5m	4
CD	Flood California flood dataset ^{EOD}	2019	0.33	RGB,MS	1	1534x808	5m,30m	2
CD	HRSCD	2019	5.06	RGB	582	10000	0.5m	7
CD	MtS-WH ^{EOD}	2019	0.459	RGB-NIR	190	150	1m	1
CD	DSIFN ^{EOD}	2020	0.454	RGB	394	512	10m	1
CD	Google Data Set ^{EOD}	2020	0.6	RGB	19	1000~5000	0.55m	2
CD	Hi-UCD ^{EOD}	2020	—	RGB	1293	1024	0.1m	9
CD	LEVIR-CD	2020	2.7	RGB	637	1024	0.5m	1
CD	S2MTcityPair ^{EOD}	2020	10.6	Multispectral (Satellite)	1520*13	600	—	1
CD	SECOND ^{EOD}	2020	2.2	RGB	4662	512	0.5~3m	30
CD	Zhang et al. CD dataset ^{EOD}	2020	0.1	RGB,NIR	4 1431x1431,452x259,158x240	1431x1431,452x259,158x240	1431x1431,452x259,158x240	240
CD	3D URB3DCD ^{EOD}	2021	1.5	—	50	/ 2 PointCloud	pm	0.5
CD	DFC21-MSD ^{EOD}	2021	325	MS,MT	2250*9	4000	1~30m	15

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Table 1 continued

Task	Dataset	Year	GB	Modality / sensor	Samples	Sample size	Resolution	Classes
CD	Dynamic EarthNet Challenge ^{EOD}	2021	–	RGB	22500	1024	3m	7
CD	HTCD ^{EOD}	2021	1.74	RGB 0.5971m,	2	11 K×15 K, 1.38 M×1.04 M	0.074m	2
CD	MUNO21 ^{EOD}	2021	14.3	Optical (Aerial)	156	12000	–	1
CD	RapidAI4EO ^{EOD}	2021	–	MS, RGB	500,000	–	3m, 10m	44
CD	Relative Radiometric Normalization ^{EOD}	2021	1	MS	7	300~5000	0.31m, 0.4m, 10m, 20m, 30m, 60m	
CD	S2Looking ^{EOD}	2021	10.2	RGB	5000*2	1024	0.5~0.8m	1
CD	S2MTCP ^{EOD}	2021	10.6	MS	1520	600	10m	–
CD	SYSU-CD ^{EOD}	2021	5.1	RGB	20000*2	256	0.5m	5
CD	WH-MAVS ^{EOD}	2021	–	RGB	47,134	200	1.2m	15
CD	CropLand CropLand Change Detection (CLCD) ^{EOD}	2022	0.5	RGB 0.5~2	600	512	m	2
CD	Dynamic World ^{EOD}	2022	–	Sentinel-2	–	–	10m	9
CD	MSBC ^{EOD}	2022	3.9	RGB, SAR, MS	3,769	256	2m	2
CD	MSOSCD ^{EOD}	2022	2.7	RGB, SAR, MS	5,107	256	10~60m	2
CD	BANDON ^{EOD}	2023	3.4	RGB	2283	2048×2048	0.6m	2
CD+C	Seasonal Contrast ^{EOD}	2021	36.3	Multispectral (Satellite)	200000*5*12	265	–	–
CD+S	DynamicEarthNet ^{EOD}	2022	524	Multispectral (Satellite)	54750	1024	–	7
CHR	Draper Satellite Image Chronology ^{EOD}	2016	36	RGB	1000	3100×2329	–	–
C	Indian Pines ^{EOD}	2000	0.006	Hyperspectral (Aerial)	1	145	20m	16
C	Salinas ^{EOD}	2000	0.027	Hyperspectral (Aerial)	1	365	3.7m	16
C	Botswana ^{EOD}	2005	0.079	Hyperspectral (Aerial)	1	875	30m	14
C	Kennedy Space Center ^{EOD}	2005	0.057	Hyperspectral (Aerial)	1	550	0.18m	13
C	Pavia Center ^{EOD}	2009	0.124	Hyperspectral (Aerial)	1	1096	–	9
C	Pavia University ^{EOD}	2009	0.033	Hyperspectral (Aerial)	1	610	–	9
C	UC Merced ^{EOD}	2010	0.317	RGB	2100	256	0.3m	21
C	WHU-RS19 ^{EOD}	2012	0.1	RGB	1013	600	0.5m	19
C	ISPRS 3D Vaihingen ^{EOD}	2014	–	ALS	780879	–	8 pts/m2	9
C	BCS ^{EOD}	2015	0.0045	Optical (Satellite)	2876	64	–	2
C	Brazilian Coffee Scene ^{EOD}	2015	0.004	RGB	2876	64	20m	2
C	NEWFOR ^{EOD}	2015	0.097	ALS	–	–	varies	4
C	RSSCN7 ^{EOD}	2015	0.086	RGB	2800	400	–	7
C	SAT-4 ^{EOD}	2015	1.15	RGB-NIR	500000	28	1~6m	4
C	SAT-6 ^{EOD}	2015	1.15	RGB-NIR	405000	28	1m	6
C	VEDAI ^{EOD}	2015	3.99	IRGB	1250	1024	0.125m	9
C	BCSS ^{EOD}	2016	0.011	Multispectral (Satellite)	1311	64	–	4
C	Brazilian Cerrado-Savanna ^{EOD}	2016	0.011	MS	1311	64	5m	4
C	NWPU-RESISC45 ^{EOD}	2016	0.405	RGB	31500	256	0.2~30m	45
C	RSC11 ^{EOD}	2016	–	RGB	1232	512	–	11
C	SIRI-WHU ^{EOD}	2016	0.7	RGB	2400	200	2m	12
C	AID ^{EOD}	2017	2.44	RGB	10000	600	3m	30
C	Austin Zoning ^{EOD}	2017	0.596	RGB	3,666	773×961	–	5
C	Golf Course MUSE ^{EOD}	2017	0.37	MS, Landsat	83431	16	30m	2
C	Kag. Planet Forest ^{EOD}	2017	33	Optical (Satellite)	150000	256	–	12

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Table 1 continued

Task	Dataset	Year	GB	Modality / sensor	Samples	Sample size	Resolution	Classes
C	RSD46-WHU ^{EOD}	2017	–	RGB	117000	256	0.5~2m	46
C	RSI-CB128 ^{EOD}	2017	0.879	RGB	36000	128	0.3~3m	45
C	RSI-CB256 ^{EOD}	2017	2.24	RGB	24000	256	0.3~3m	35
C	Ships in Satellite Imagery ^{EOD}	2017	0.343	RGB	4000	80	3m	2
C	Solar Power Plants MUSIC4P3 ^{EOD}	2017	4.6	MS,Landsat	1280000	16	30m	2
C	TiSeLaC ^{EOD}	2017	–	RGB-NIR,MT	23	2866x2633	30m	9
C	DeepGlobe-Road ^{EOD}	2018	3.84	Optical (Satellite)	8570	1024	–	1
C	EuroSAT	2018	1.92	Sentinel-2	27000*13	64	10m	10
C	FMoW	2018	3,500	RGB,MS	523846	–	0.3m	63
C	Gaofen Image (GID)	2018	47	RGB-NIR	150	7200	4m	15
C	MASATI-v2	2018	2.3	RGB	7389	512	0.08~2m	7
C	MSLCC ^{EOD}	2018	0.5	SAR,MS	2	5596x6031,8149x31057	10m	4
C	PatternNet	2018	1.3	RGB	30400	256	0.062~4.693m	38
C	SanFranBay Ships Planet ^{EOD}	2018	0.194	Optical (Satellite)	4000	80	–	1
C	xView	2018	20	RGB,MS	1413	3000	0.3m	60
C	AIDER ^{EOD}	2019	0.275	Optical (Drone)	2645	240	–	4
C	BigEarthNet-MM ^{EOD}	2019	121	Sentinel-1, Sentinel-2 10m, 20m,	590326*12	120	60m	19
C	DSCR	2019	–	RGB	20,675	150~800	–	7
C	HistAerial ^{EOD}	2019	7.6	Grayscale	42000	25,50,100	–	7
C	ICONES-HSI ^{EOD}	2019	11.5	Hyperspectral (Aerial)	468*224	300	–	9
C	Kag. Find a Car Park ^{EOD}	2019	2.75	RGB	3262	1296	–	2
C	Kag. Satellite Hurricane Damage ^{EOD}	2019	0.064	RGB	16000	128	1m	–
C	Kaggle Cloud Detection ^{EOD}	2019	6	RGB	9244	1750	–	4
C	OPTIMAL-31 ^{EOD}	2019	0.024	RGB	1860	256	–	31
C	Ship Classification ^{EOD}	2019	–	Optical (Satellite)	1951	–	–	7
C	Slovenia Land Cover ^{EOD}	2019	11.55	Sentinel-2	940	500	10m	10
C	SloveniaLandCover ^{EOD}	2019	190	Multispectral (Satellite)	940	500	–	10
C	So2Sat LCZ42 ^{EOD}	2019	51.8	Sentinel-1, Sentinel-2	400673	32	10m	–
C	USTC SmokeRS ^{EOD}	2019	0.795	RGB	6225	256	1000m	6
C	WiDS Datathon 2019 ^{EOD}	2019	0.46	RGB	11000	256	3m	2
C	AiRound	2020	33	RGB, Sentinel-2, Ground, Aerial	11753	300	–	11
C	Airbus Wind Turbines Patches ^{EOD}	2020	1	RGB,MS	358000	128	1.5m	1
C	ArCH ^{EOD}	2020	–	Multiple	136138423	–	varies	10
C	Bijie Landslide	2020	0.513	RGB	2773	200	0.68m	1
C	BrazilDAM	2020	57	RGB	769	384	10~60m	2
C	BreizhCrops ^{EOD}	2020	–	Sentinel-2, MT	610000	–	10~60m	9
C	CLRS ^{EOD}	2020	1.74	RGB	15000	256	0.26~8.85m	25
C	CV-BrCT	2020	9.2	RGB	24000	500	–	9
C	CV4A Kenya ^{EOD}	2020	3.5	Sentinel-2	8788	2500	10m	7
C	CaneSat ^{EOD}	2020	0.006	Sentinel-2, MT	1627	10	10m	2
C	DND-SB ^{EOD}	2020	49	Optical (Drone)	5648	7296	–	7
C	LandCoverNet Af ^{EOD}	2020	632	Multiple (Satellite)	1980	256	–	7
C	LandCoverNet As ^{EOD}	2020	610	Multiple (Satellite)	2753	256	–	7
C	LandCoverNet Au ^{EOD}	2020	126	Multiple (Satellite)	600	256	–	7

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Table 1 continued

Task	Dataset	Year	GB	Modality / sensor	Samples	Sample size	Resolution	Classes
C	LandCoverNet ^{EOD} _{Eu}	2020	249	Multiple (Satellite)	840	256	–	7
C	LandCoverNet ^{EOD} _{NA}	2020	351	Multiple (Satellite)	1561	256	–	7
C	LandCoverNet ^{EOD} _{SA}	2020	215	Multiple (Satellite)	1200	256	–	7
C	MLRSNet ^{EOD}	2020	1.25	RGB	109161	256	0.1~10m	46
C	OVERHEAD MNIST ^{EOD}	2020	0.018	Grayscale	1000	28	–	9
C	Rwanda Crop Type ^{EOD}	2020	56	Optical (Drone)	6450	200	–	6
C	SAR-based Sea Ice Dataset ^{EOD}	2020	0.14	SAR (Satellite)	98754	28	–	6
C	Sen1Floods11	2020	14.3	Sentinel-1	4831	512	10m	1
C	SenseEarth classify ^{EOD}	2020	10.8	RGB	70000	100~12655	0.2~153m	51
C	2019 Mali Crop-Type ^{EOD}	2021	0.456	Multispectral (Satellite)	33*13*41	256	–	4
C	CanCropland ^{EOD}	2021	–	Multispectral (Satellite)	78536*15	64	–	10
C	CropRotation ^{EOD}	2021	53.4	Multispectral (Satellite)	–	–	–	–
C	DENETHOR ^{EOD}	2021	254	Sentinel-1, Sentinel-2	2*(365 + 60*14)	8000	3m	9
C	FGSCR-42 ^{EOD}	2021	–	RGB	9320	500	–	42
C	MAFAT-Fine-Grained ^{EOD}	2021	–	RGB 0.05~	4216	–	0.15m	37
C	Million AID ^{EOD}	2021	133.5	RGB	1,000,000	150~550	0.5~153m	28
C	MultiScene ^{EOD}	2021	6	RGB	100000	512	0.3~0.6m	36
C	NaSC-TG2 ^{EOD}	2021	–	RGB-NIR	20000	256	100m	10
C	Satellite Image Classification ^{EOD}	2021	0.023	RGB	5631	256	–	4
C	SeaIceClassification ^{EOD}	2021	2.9	SAR (Satellite)	6400	256	–	8
C	South Africa Crop Type ^{EOD}	2021	82.77	Sentinel-1, Sentinel-2	122736	–	–	9
C	TG1HRSSC ^{EOD}	2021	0.277	Hyperspectral 5m, 10m,	204	512	20m	9
C	Urban Village S2UC ^{EOD}	2021	1.8	RGB	1714	224	2m	2
C	ZueriCrop	2021	38.5	Sentinel-2	28000*71	24	10m	48
C	ADVANCE ^{EOD}	2022	4.4	Optical (Satellite)	5075	256	–	13
C	ARGO ^{EOD}	2022	14.3	Multispectral (Satellite)	1750	1400	–	2
C	AgriFieldNet ^{EOD}	2022	0.32	Multispectral (Satellite)	1217*12	256	–	13
C	Aircraft SAR-ACD ^{EOD}	2022	–	SAR	4322	64	–	20
C	AnthroProtect ^{EOD}	2022	18.1	Optical (Satellite)	23919	256	–	2
C	Hot Area MUISIC4HA (Hot Area) ^{EOD}	2022	0.01	Sentinel-2	2511	16	10m	6
C	OMBRIA ^{EOD}	2022	0.19	Sentinel-1, Sentinel-2	3376	256	10m~20m	2
C	S2-Agri ^{EOD}	2022	10.8	Multispectral (Satellite)	–	32	–	–
C	Space2Ground ^{EOD}	2022	0.501	Sentinel-1, Sentinel-2, RGB	10102	260	10~60m	2
C	SynthWakeSAR ^{EOD}	2022	4.3	SAR	46080	96000	3.3m	10
C	The Auto Arborist Dataset ^{EOD}	2022	24	RGB, MS	2,637,208	1,024	–	344
C	The Canadian Crop-land ^{EOD}	2022	26	Sentinel-2	78536	64	10m	10
C	TreeSatAI ^{EOD}	2022	16.3	Sentinel-1, Sentinel-2, RGB	50381	304	10m, 0.2m	47

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Table 1 continued

Task	Dataset	Year	GB	Modality / sensor	Samples	Sample size	Resolution	Classes
C	XAI4SAR ^{EOD}	2022	0.691	SAR (Satellite)	5915	256	–	–
C	FireRisk ^{EOD}	2023	14	Optical (Satellite)	91872	–	–	7
C	MAR20 ^{EOD}	2023	1.1	RGB	3842	800×800	1m	20
C+O	CropHarvest ^{EOD}	2021	22.7	Sentinel-1, Sentinel-2, ERA5, DEM	95186	–	10~60m	–
CLIM	Earth Surface Temperature ^{EOD}	2017	0.09	Climate variables	16	–	–	–
CLIM	Greenhouse Gas ^{EOD}	2017	0.0001	Gas emission	–	–	–	–
CR	SEN12MS-CR	2020	272	MS, SAR	122218	256	10m	–
CR	SEN12MS-CR-TS ^{EOD}	2021	649	MS, SAR	53	4000	10m	–
CR	Planet-CR (M3R-CR) ^{EOD}	2022	–	Multiple (Satellite)	63000	300	–	–
DH	SateHazelk ^{EOD}	2017	1.2	RGB	1200	512	3m	–
FLT	OpenGF ^{EOD}	2021	2.28	ALS	542100000	–	6 and 14 pts/m2	3
FOR	Forest Canopy Height ^{EOD}	2018	62.9	LiDAR	1105	–	1m	–
GL	CVUSA ^{EOD}	2017	15.7	RGB	44416	750x750	0.3m	0
GL	UCF Cross View Dataset ^{EOD}	2017	59.5	RGB	35404	1200x1200	–	0
GL	CVACT ^{EOD}	2019	152.8	RGB	128334	1200x1200	0.5m	0
GL	University-1652 ^{EOD}	2020	–	RGB	146580	–	–	–
GL	VIGOR ^{EOD}	2021	94.2	RGB	90618	640x640	0.114m	0
GEO	TenGeoP-SARwv ^{EOD}	2019	31.7	SAR	37000	–	5m	10
IC	Sydney caption ^{EOD}	2016	0.298	RGB	613	500	0.5m	7
IC	UCM caption ^{EOD}	2016	0.255	RGB	2100	256	0.3m	21
IC	LEVIR-CC ^{EOD}	2022	–	RGB	10077	512	–	–
IM	EuroSDR Image Matching ^{EOD}	2014	–	RGB	–	–	6-13m	–
IR	Dataset of thermal and visible ^{EOD}	2019	0.31	RGB, Thermal	110	336, 4000	–	0
IT	Aerial to Map (Pixel2Pixel) ^{EOD}	2017	0.239	RGB	2194	600	–	–
IT	WHU-SEN-City ^{EOD}	2019	4.3	Sentinel-1, Sentinel-2	18542	256	10m	–
M3D	The IARPA Multi-View Stereo 3D ^{EOD}	2017	75	MS, LiDAR	–	–	0.3m	–
M3D	WHU MVS/Stereo Dataset ^{EOD}	2020	95.7	RGB	1776	5376	0.1m	–
M3D	WHU Multi-View Dataset ^{EOD}	2020	12.3	RGB	28400	768	0.1m	–
M3D	WHU Stereo Dataset ^{EOD}	2020	8.4	RGB	21868	768	0.1m	–
M3D	WHU TCL Sat-MVS ^{EOD}	2021	29	Panchromatic	300	5120	2.1-2.5m	–
OD	MSTAR ^{EOD}	1996	–	SAR (Satellite)	–	–	0.3m	–
OD	SAR Vehicle ^{EOD}	2005	7.5	SAR (Aerial)	104	–	–	1
OD	VIVID - PETS2005 ^{EOD}	2007	–	Optical (Aerial)	9283	640	–	1
OD	TaS ^{EOD}	2008	0.97	RGB	30	792	0.5m	1
OD	OIRDS ^{EOD}	2009	0.153	RGB	900	256~640	0.15m	5
OD	SZTAKI AirChange	2012	0.042	RGB	13	800	1.5m	2
OD	Massachusetts Roads	2013	7.55	RGB	1171	1500	1m	–
OD	NWPU-VHR10	2014	0.073	RGB, IRRG	800	1000	0.08~2m	10
OD	UCAS_AOD ^{EOD}	2014	3.24	RGB	976	1,000	–	2
OD	PKLot ^{EOD}	2015	4.6	RGB	12417	1280	UAV	1
OD	CARPK ^{EOD}	2016	2	RGB	1448	1280	UAV	1
OD	COWC	2016	64	RGB	388435	256	0.15m	1
OD	DLR3k/DLR-MVDA ^{EOD}	2016	0.162	RGB	20	3744	0.13m	7

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Table 1 continued

Task	Dataset	Year	GB	Modality / sensor	Samples	Sample size	Resolution	Classes
OD	DSTL16B ^{EOD}	2016	7.3	Multispectral (Satellite)	1350	3391	–	10
OD	DSTL3B ^{EOD}	2016	12.9	Optical (Satellite)	450	3391	–	10
OD	HSI Vehicle ^{EOD}	2016	–	Hyperspectral (Aerial)	157	–	–	1
OD	LEVIR ^{EOD}	2016	–	RGB	21952	800	0.5m	3
OD	Stanford Drone Dataset ^{EOD}	2016	69	RGB Video	–	–	0.025m	6
OD	HRSC2016 ^{EOD}	2017	8	RGB	1061	1100	0.4~2m	26
OD	Land Covers Dstl Satellite Imagery ^{EOD}	2017	21.7	RGB,MS	57	3,348	0.3m~7.5m	10
OD	NOAA Sea Lion Count ^{EOD}	2017	96	RGB	950	4,900	–	4
OD	OpenSARShip ^{EOD}	2017	0.87	Sentinel-1	11346	–	10m	17
OD	OpenSARShip 2.0 ^{EOD}	2017	0.87	Sentinel-1	34528	120	10m	17
OD	PlanesNet ^{EOD}	2017	0.39	Optical (Satellite)	32000	20	–	–
OD	RSOD	2017	0.309	RGB	946	1000	0.3~3m	4
OD	SSDD ^{EOD}	2017	0.995	SAR (Satellite)	1160	500	1~15m	1
OD	WAMI DIRSIG ^{EOD}	2017	0.327	Hyperspectral (Aerial)	55226	64	0.3m	1
OD	DOTA v1.0	2018	18	RGB	2806	4000	–	15
OD	DeepGlobe (Building) ^{EOD}	2018	–	Panchromatic,RGB,MS	24586	650	0.5m	2
OD	EU Flood ^{EOD}	2018	5.6	Optical (Multiple)	3710	1280	–	4
OD	HighD ^{EOD}	2018	–	RGB	110000	4096x2160	–	2
OD	ITCVD ^{EOD}	2018	1.3	RGB	173	5616	0.1m	1
OD	Iceberg Detection ^{EOD}	2018	0.29	SAR (Satellite)	10028	75	–	–
OD	Kag. Airbus Ship Detection	2018	30	RGB	192556	768	–	–
OD	Kaggle Airbus Ship ^{EOD}	2018	31.41	–	192556	768 2	1.5m	–
OD	Map Challenge ^{EOD}	2018	–	RGB	341,058	300	–	2
OD	Ships in Google Earth ^{EOD}	2018	2	RGB	794	2000	–	2
OD	SpaceNet-4	2018	186	MS,WorldView2	60000	900	0.3m	–
OD	TBF ^{EOD}	2018	–	RGB	13	40,000	–	2
OD	WHU Building	2018	25	RGB	25577	512	0.3m	1
OD	AIR-SARShip-1.0 ^{EOD}	2019	0.2	SAR (Satellite)	31	3000	1~3m	1
OD	ALCD ^{EOD}	2019	0.234	MS,Sentinel-2	38	1830	10~60m	–
OD	Bridges	2019	1.45	RGB	500	3822	0.5m	1
OD	DLR-ACD ^{EOD}	2019	–	Optical (Aerial)	33	4458	–	1
OD	DOTA v1.5	2019	18	RGB	2806	4000	–	16
OD	Interaction Dataset ^{EOD}	2019	–	RGB	10,933	–	–	1
OD	MTARSI ^{EOD}	2019	0.48	RGB	9385	256	–	1
OD	Microsoft Building (Australia) ^{EOD}	2019	6.4	–	11,334,866	/ 2	–	–
OD	NYCPlanimetric ^{EOD}	2019	2	Optical (Aerial)	2147	5000	–	24
OD	Oil Palm Plant ^{EOD}	2019	0.5	RGB	20,000	–	3m	2
OD	Oil Storage Tanks	2019	3	RGB	10000	512	0.5m	1
OD	SAR Ship Detection	2019	0.407	SAR 3m, 5m,	43819	256	8m,10m	1
OD	SAR VD ^{EOD}	2019	7.5	SAR (Aerial)	104	980	–	1
OD	SIMD ^{EOD}	2019	1	RGB	5000	1024	UAV@150m	15
OD	Swimming pool/car ^{EOD}	2019	0.12	RGB	3750	224	–	2
OD	VisDrone2019-DET ^{EOD}	2019	2	RGB	10209	2000x1500	UAV	10

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Table 1 continued

Task	Dataset	Year	GB	Modality / sensor	Samples	Sample size	Resolution	Classes
OD	VisDrone2019-MOT ^{EOD}	2019	14	RGB Video	40000	3840x2160	UAV	5
OD	VisDrone2019-SOT ^{EOD}	2019	68	RGB Video	139300	3840x2160	UAV	3
OD	VisDrone2019-VID ^{EOD}	2019	14	RGB	40000	3840x2160	UAV	5
OD	built-structure-count ^{EOD}	2019	2	RGB	5364	512	0.3m	–
OD	AI-TOD ^{EOD}	2020	22	RGB	28036	–	0.3m~30m	–
OD	AIR-SARShip-2	2020	0.224	SAR (Satellite)	300	1000	1~3m	1
OD	AU-AIR	2020	2.2	RGB	32823	1920	UAV@30m	8
OD	Aerial Maritime Drone ^{EOD}	2020	0.038	RGB	508	800x600	–	5
OD	Agriculture-Vision	2020	4.39	Hyperspectral (Aerial)	94986	512	0.463m	6
OD	BIRDSAI	2020	43.2	Thermal	162000	640	UAV@60-120m	–
OD	DOTA v2.0	2020	34.3	RGB	11268	4000	0.1~0.81	18
OD	DroneRGBT ^{EOD}	2020	–	Multiple (Drone)	3600	640	–	1
OD	EAGLE ^{EOD}	2020	–	RGB	8820	936	0.05~0.45m	–
OD	EoC ^{EOD}	2020	0.217	Optical (Aerial)	2800	256	–	14
OD	FAIR1M ^{EOD}	2020	120	Optical (Satellite)	15266	5000	–	37
OD	FGSD ^{EOD}	2020	–	Optical (Satellite)	4736	930	–	43
OD	HRSID	2020	0.581	Sentinel-1,SAR	5604	800	0.5~3m	1
OD	Intersection Drone ^{EOD}	2020	–	RGB	11,500	4096x2160	–	5
OD	LS-SSDD	2020	7.8	SAR (Satellite)	15	20000	1~15m	1
OD	MOR-UAV ^{EOD}	2020	–	RGB	10948	608	UAV	2
OD	Oil and Gas Tank (OGST) ^{EOD}	2020	1.87	RGB	10000	512	0.3m	2
OD	RSOC ^{EOD}	2020	–	RGB	3057	2500	–	4
OD	RarePlanes	2020	318	MS,WorldView3	713348	512	0.3~1.5m	110
OD	Roundabouts Drone ^{EOD}	2020	–	RGB	13,746	4096x2160	–	8
OD	TGRS-HRRSD ^{EOD}	2020	9	RGB 0.15~1.2	21761	930	m	13
OD	Tree NEON Tree Crowns ^{EOD}	2020	27.4	RGB	11,000	100 million trees	–	2
OD	VisDrone	2020	85	RGB	275437	1400	UAV	–
OD	AFO ^{EOD}	2021	5	RGB	3647	3840	–	2
OD	AMD ^{EOD}	2021	0.037	Optical (Drone)	508	800	–	5
OD	Airbus Aircraft Detection ^{EOD}	2021	0.092	RGB	103	2560	0.5m	1
OD	Airbus Oil Storage Detection ^{EOD}	2021	0.101	MS	98	2560	1.2m	1
OD	AnimalDrone ^{EOD}	2021	–	Optical (Drone)	53644	–	–	–
OD	ArcticSeals ^{EOD}	2021	1,000	Multispectral (Aerial)	88000	6576	–	1
OD	ArtifiVe-Potsdam ^{EOD}	2021	15.6	MS	4800	600	0.05m	1
OD	CASIA-Ship ^{EOD}	2021	–	RGB	1,118	1,680	–	2
OD	CASIA-aircraft ^{EOD}	2021	–	RGB	58,121	399	–	2
OD	CGI Planes ^{EOD}	2021	0.7	RGB	500	–	–	2
OD	DSSDD ^{EOD}	2021	1.27	SAR (Satellite)	1236	256	–	1
OD	DroneVehicle ^{EOD}	2021	–	RGB-Infrared	56878	840	UAV@100m	5
OD	Flying Airplanes ^{EOD}	2021	491	Optical (Satellite)	8000	–	–	1
OD	Forest Damages ^{EOD}	2021	3.3	RGB	1543	1500	UAV	5
OD	LEVIR-Ship ^{EOD}	2021	0.821	Optical (Satellite)	3896	512	–	1
OD	MultiDrone ^{EOD}	2021	–	Optical (Drone)	259793	1280	–	9
OD	NASADEbris ^{EOD}	2021	0.073	Optical (Satellite)	707	256	–	1
OD	SRSD-v1.0 ^{EOD}	2021	0.713	SAR (Satellite)	666	1024	–	6

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Task	Dataset	Year	GB	Modality / sensor	Samples	Sample size	Resolution	Classes
OD	SWIM ^{EOD}	2021	12	RGB	14610	768	0.5~2.5m	1
OD	SaRNet ^{EOD}	2021	0.894	Optical (Satellite)	2552	1000	–	1
OD	SeabirdsDetection ^{EOD}	2021	2.2	Optical (Aerial)	1	28814	–	1
OD	Search/Rescue Search And Rescue ^{EOD}	2021	–	RGB	2552	1000	0.5m	1
OD	ShipRSImageNet ^{EOD}	2021	4.3	Optical (Satellite)	3435	930	–	1
OD	TopoBoundary ^{EOD}	2021	20	Optical (Aerial)	25297	1000	–	8
OD	Urban Tree Crown Detection ^{EOD}	2021	0.2	Optical (Aerial)	220	512	–	1
OD	WHUS2-CD+ ^{EOD}	2021	27.8	Sentinel-2	36	10980	10m	–
OD	Yosemite Tree ^{EOD}	2021	2.7	Optical (Aerial)	1	38400	–	1
OD	xView3-SAR ^{EOD}	2021	2,000	Sentinel-1	991*2	27000	10m	2
OD	AISD ^{EOD}	2022	–	RGB	–	–	–	–
OD	BONAI ^{EOD}	2022	4.86	RGB	3,300	1,024	0.3m~0.6m	1
OD	Buildings Containers Cranes ^{EOD}	2022	3	Optical (Satellite)	–	2500	–	3
OD	DroneCrowd ^{EOD}	2022	10.4	Optical (Drone)	33600	1920	–	1
OD	HRPlanesv2 ^{EOD}	2022	8.1	Optical (Satellite)	2120	4800	–	1
OD	MSAR ^{EOD}	2022	2.5	SAR (Satellite)	28449	256	–	4
OD	OnlyPlanes ^{EOD}	2022	15	Optical (Satellite)	12500	1024	–	1
OD	RSDD-SAR ^{EOD}	2022	–	SAR (Satellite)	7000	512	–	–
OD	RampBuilding ^{EOD}	2022	24.8	Optical (Satellite)	96284	256	–	1
OD	SADD ^{EOD}	2022	–	SAR (Satellite)	2966	224	–	1
OD	SAR Aircraft ^{EOD}	2022	0.18	SAR	2966	224	0.5m~3m	2
OD	SODA-A ^{EOD}	2022	–	RGB	2510	4761×2777	–	9
OD	Satellite Pool Detection ^{EOD}	2022	3.7	Optical (Satellite)	14912	512	–	1
OD	SeaDronesSee ^{EOD}	2022	60.3	RGB	5630	3,840~5,456	–	6
OD	UAVOD-10 ^{EOD}	2022	0.902	Optical (Drone)	844	4800	–	10
OD	UAVOD10 ^{EOD}	2022	0.9	RGB	844	1000~4800	0.15m	10
OD	Urban Building Classification ^{EOD}	2022	0.675	RGB	800	600	0.5~0.8m	61
OD	VISO ^{EOD}	2022	8.6	Optical (Satellite)	15455	12000	–	4
OD	Volcanoes Hephaestus ^{EOD}	2022	93.71	InSAR	216106	224	–	6
OD	MDMT ^{EOD}	2023	25.8	Optical (Drone)	39678	–	–	–
OD	DIOR ^{EOD}	–	–	RGB 0.5~30	23463	800	m	20
OD	MSDS900 ^{EOD}	–	–	Optical (Satellite)	2993	900	–	1
OD	MSDSfull ^{EOD}	–	–	Optical (Satellite)	200	4800	–	1
OD	PSDS ^{EOD}	–	–	Optical (Satellite)	1310	900	–	1
OD	SMCDD ^{EOD}	–	–	SAR (Satellite)	–	–	–	–
OD+CNOAA	Sea Lion ^{EOD}	2017	96	RGB	950	4900	–	4
OD+CHumpback	Whale Identification ^{EOD}	2018	0.7	Optical (Multiple)	25460	1050	–	–
OD+CFUSAR-Ship	^{EOD}	2020	–	SAR (Satellite)	5000	512	–	113
OD+CIDTReeS	^{EOD}	2020	0.28	Multiple (Multiple)	591*369	20	–	33
OD+CForest	Damages – Larch Case-bearer ^{EOD}	2021	3.3	RGB	1543	1500	UAV	4
OD+CRefores	Tree ^{EOD}	2022	7.5	Optical (Drone)	100	4000	–	6
OD+ONEON	Tree Dataset ^{EOD}	2021	8.4	Multiple (Aerial)	2174*426	40	–	1
OD+SVALID	^{EOD}	2020	15.7	RGBD	6690	1024	–	30

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Task	Dataset	Year	GB	Modality / sensor	Samples	Sample size	Resolution	Classes
OD+SWHU	Cloud	2020	3.65	Multispectral (Satellite)	859	512	–	1
OD+S	LiSurveying ^{EOD}	2021	–	TLS	2450000000	–	varies	54
O	HighDenseMatching ^{EOD}	2014	72	Optical (Aerial)	186	15000	–	–
O	DFC16 ^{EOD}	2016	–	Optical (Satellite)	–	–	–	–
O	SPARCS ^{EOD}	2016	1.4	MS,Landsat	80	1000	30m	–
O	GTA BEV SVA ^{EOD}	2017	350	Optical (Satellite)	1000000	1920	–	–
O	RSICD	2017	0.459	Optical (Satellite)	10921	224	–	–
O	RoadTracer ^{EOD}	2017	0.083	RGB	40	8150	0.6m	1
O	RoofN3D ^{EOD}	2018	–	ALS	118100	–	4.72 pts/m2	3
O	DFC19 ^{EOD}	2019	–	MS,LiDAR	–	–	0.3~1.3m	–
O	ForestCanopyHeight ^{EOD}	2019	62.9	Multiple (Aerial)	1105	–	–	–
O	MidAir ^{EOD}	2019	1,000	RGBD,Odometry	420000	1024	–	–
O	SPEED ^{EOD}	2019	4.9	RGB	15300	1920	0.5m	–
O	SatStereo	2019	127	Optical (Satellite)	144	–	–	–
O	ALPACA ^{EOD}	2020	37	Hyperspectral (Drone)	–	–	–	–
O	CloudCast ^{EOD}	2020	328	NWP	70080	1229	3km	–
O	ERA	2020	6.3	RGB Video	343680	640	–	–
O	GeoLifeCLEF ^{EOD}	2020	840	RGB-IR,MS,LC,DEM	6700000	256	1m,0.3m,0.1m	–
O	GeoLifeCLEF ^{EOD}	2020	–	Multiple (Multiple)	–	–	–	–
O	RSVQA-HR ^{EOD}	2020	14.4	Optical (Aerial)	10659	512	–	–
O	RSVQA-LR ^{EOD}	2020	0.15	Optical (Satellite)	772	256	–	–
O	Satellite Pose Estimation ^{EOD}	2020	4.6	Optical (Satellite)	15300	1920	–	–
O	WHU MVS/Stereo ^{EOD}	2020	98	Optical (Aerial)	1776	5376	–	–
O	WHU Multi-View ^{EOD}	2020	12.6	Optical (Aerial)	28400	768	–	–
O	WHU Stereo ^{EOD}	2020	8.6	Optical (Aerial)	21868	768	–	–
O	Africa MiniCubes ^{EOD}	2021	47	–	–	–	–	–
O	DLR-HySU ^{EOD}	2021	0.005	Hyperspectral (Aerial)	1*135	123	–	–
O	DREAM ^{EOD}	2021	68	Multiple (Multiple)	–	1024	–	–
O	DenseMatchAerial ^{EOD}	2021	3.3	Optical (Aerial)	1092*2	1024	–	–
O	EarthNet2021 ^{EOD}	2021	600	Multispectral (Satellite)	32000	128	–	–
O	ExtremeEarth ^{EOD}	2021	0.246	–	–	–	–	–
O	GeoLifeCLEF ^{EOD}	2021	–	RGB-IR,MS,LC,DEM	–	–	1m,0.3m,0.1m	–
O	Hurricane Wind Speed ^{EOD}	2021	1.5	Multispectral (Satellite)	114634	366	–	–
O	MultimodalSupCD ^{EOD}	2021	0.66	Multiple (Satellite)	–	96	–	–
O	QXS-SAROPT ^{EOD}	2021	2.7	Multiple (Satellite)	40000	256	–	–
O	RSITMD ^{EOD}	2021	0.568	Optical (Satellite)	4743	–	–	–
O	RSVQAxBEN ^{EOD}	2021	28.6	Optical (Satellite)	590326	120	–	–
O	SPYSTUF ^{EOD}	2021	29	Hyperspectral (Aerial)	1*128	16000	–	–
O	SeaIceContent ^{EOD}	2021	0.245	SAR (Satellite)	6	25600	–	–
O	SustainBench ^{EOD}	2021	302	Multiple (Multiple)	–	–	RGB,Time-series	–

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Task	Dataset	Year	GB	Modality / sensor	Samples	Sample size	Resolution	Classes
O	TIRMSS ^{EOD}	2021	8.45	Multispectral (Satellite)	16000	192	—	—
O	AI4Arctic Sea Ice ^{EOD}	2022	—	SAR (Satellite)	—	—	—	—
O	Hephaestus ^{EOD}	2022	105	InSAR	110573	224	—	—
O	MDAS ^{EOD}	2022	35	Multiple (Multiple)	—	—	—	—
O	PLP ^{EOD}	2022	160	Multiple (Multiple)	—	—	—	—
O	Phenology ^{EOD}	2022	56	Optical (Satellite)	114*24*5	3333	—	—
O	SAR2Opt ^{EOD}	2022	0.691	Multiple (Satellite)	—	600	—	—
O	SSL4EO-S12	2022	1,500	Multiple (Satellite)	251079	264	—	—
O	SyntCities ^{EOD}	2022	38.2	Optical (Multiple)	16200	1024	—	13
O	UAE-RS ^{EOD}	2022	1.01	Optical (Multiple)	6050	500	—	—
O	SEN12TP ^{EOD}	2023	222	Multiple (Satellite)	2319	—	—	—
O	mesogeos ^{EOD}	2023	—	—	—	—	—	—
O	MAFAT Challenge ^{EOD}	—	—	—	—	—	—	—
O	Olive tree ^{EOD}	—	—	—	10	4000	—	—
O	So2Sat POP ^{EOD}	—	40	—	—	—	—	—
PT	Cactus Aerial Photos ^{EOD}	2018	0.053	RGB	24000	32	—	2
PT	Urban Tree Detection Data ^{EOD}	2020	0.114	RGB-NIR	60	256	0.6m	2
POP	So2Sat Population ^{EOD}	2022	—	DEM, Sentinel-2	—	—	10m	—
POV	Poverty in Africa ^{EOD}	2020	5.58	RGB	32823	256	—	—
R	Tropical Cyclone	2020	—	GOES 8–16 MSI	108110	256x256	4K–8K	—
R	USAVars	2021	—	NAIP RGB+NIR	100K	—	4m	—
SSL	SeCo ^{EOD}	2021	43.6	Sentinel-2	24000x5	264	10m	—
S	DFC07 ^{EOD}	2007	0.019	Multiple (Satellite)	1	787	—	—
S	DFC08 ^{EOD}	2008	0.194	Hyperspectral (Aerial)	—	—	—	5
S	ISPRS 2D - Potsdam ^{EOD}	2011	16	RGB,nDSM	38	6000	0.05m	6
S	ISPRS 2D - Vaihingen ^{EOD}	2011	17	RGB,nDSM	33	2200	0.09m	6
S	Aerial Image Segmentation ^{EOD}	2013	0.007	RGB	80	512	0.3~1m	—
S	DFC13 ^{EOD}	2013	1	Multiple (Aerial)	145	—	—	16
S	DFC14 ^{EOD}	2014	—	Multiple (Aerial)	—	—	—	7
S	DFC15 ^{EOD}	2015	—	Multiple (Aerial)	7	10000	—	—
S	DFC2015 Zeebruges ^{EOD}	2015	0.0024	RGB,DSM,LiDAR	7	100,000	0.05m	8
S	ERM PAIW ^{EOD}	2015	0.635	RGB	41	4000	0.3m	2
S	Zurich Summer	2015	0.132	RGB-NIR	20	1000	0.61m	—
S	Biome: L8 Cloud Cover ^{EOD}	2016	96	RGB	96	—	30m	4
S	Chikusei ^{EOD}	2016	1.4	Hyperspectral (Aerial)	1*128	2517	—	19
S	HD-Maps ^{EOD}	2016	0.146	RGB	20	4000	0.3m	5
S	SpaceNet-1	2016	31	RGB,MS	9735	650	0.5~1m	1
S	AIS From Online Maps ^{EOD}	2017	23.8	RGB	1671	3000	0.5m	2
S	CITY-OSM ^{EOD}	2017	23.8	Optical (Aerial)	21	2500	—	—
S	DFC17 ^{EOD}	2017	—	Multiple (Multiple)	—	—	—	17

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Table 1 continued

Task	Dataset	Year	GB	Modality / sensor	Samples	Sample size	Resolution	Classes
S	EvLab-SS ^{EOD}	2017	–	RGB	60	4500	0.1m,0.25m	11
S	INRIA	2017	19.5	RGB	360	1500	0.3m	1
S	Semantic 3D ^{EOD}	2017	23.94	TLS	4000000000	–	–	8
S	SpaceNet-2	2017	182	RGB,MS	24586	650	0.3m	1
S	SpaceNet-3	2017	182	Panchromatic,RGB,MS	3711	1300	0.3~1.24m	1
S	38-Cloud	2018	16	RGB	17601	384	30m	1
S	AerialLanes18 ^{EOD}	2018	–	RGB	20	5616	0.125m	–
S	Aeroscapes ^{EOD}	2018	0.752	RGB	3269	1280	UAV@5-50m	11
S	Agricultural Crop Cover ^{EOD}	2018	4.4	MS	40	–	30m	2
S	CrowdAI Mapping	2018	5.33	RGB	401755	300	–	1
S	DFC18 ^{EOD}	2018	19.8	MS,Hyperspectral,RGB	1	–	0.05~1m	20
S	DLRSD ^{EOD}	2018	0.004	RGB	2100	256	0.3m	21
S	DeepGlobe-LandCover ^{EOD}	2018	2.75	RGB	1146	2448	0.5m	7
S	HyRANK ^{EOD}	2018	0.4	Hyperspectral	5	1,000	30m	14
S	RIT-18 ^{EOD}	2018	1.5	MS	1	9300	0.047m	–
S	RoadNet ^{EOD}	2018	0.905	RGB	20	–	0.21m	2
S	SEN1-2 ^{EOD}	2018	43.7	Multiple (Satellite)	282384	256	–	–
S	SynthAer ^{EOD}	2018	1	RGB	765	1280	–	8
S	Urban Drone Dataset (UDD) ^{EOD}	2018	1.1	RGB	301	4,096	UAV	6
S	WHDLD ^{EOD}	2018	0.102	RGB	4940	256	2m	6
S	AIRS	2019	17.6	RGB	1047	10000	0.075m	1
S	AeroRIT ^{EOD}	2019	1.8	RGB,Hyperspectral	1	3900	0.4m	–
S	Chesapeake Land Cover ^{EOD}	2019	140	RGB,MS	250000	240	1m	16
S	DLR-SkyScapes ^{EOD}	2019	–	RGB	16	4680	0.13m	31
S	DroneDeploy ^{EOD}	2019	–	RGB	55	6,000	0.1m	7
S	GF2 Dataset for 3DFGC ^{EOD}	2019	0.056	RGB-NIR	11	2,652	4m	5
S	HRC_WHU ^{EOD}	2019	3.56	RGB	150	1280x720 2 RGB 0.5~15m 0.17 WHU Cloud Dataset[235] 2020 859 512	30m	2
S	ManipalUAVid ^{EOD}	2019	1.05	Optical (Drone)	711	1280	–	4
S	Open AI Challenge: Caribbean ^{EOD}	2019	–	RGB	7	52,318	0.04m	5
S	PolSF ^{EOD}	2019	–	SAR (Satellite)	5	–	–	6
S	STGAN Cloud Removal	2019	1.5	Optical (Satellite)	217190	256	–	1
S	Salient Objects ORSSD ^{EOD}	2019	0.026	RGB	800	500	–	8
S	Semantic Drone ^{EOD}	2019	4	RGB	400	5000	–	20
S	Semantic Drone Dataset-SemSeg ^{EOD}	2019	3.91	RGB	400	5000	–	20
S	SpaceNet-5	2019	84.1	Panchromatic,RGB,MS	2369	1300	0.3m	1
S	95-Cloud	2020	18	RGB	34701	384	30m	1
S	AIR-CD ^{EOD}	2020	12	RGB-NIR	34	7300	4m	1
S	ActiveFireL8 ^{EOD}	2020	546	Multispectral (Satellite)	2000000*10	128	–	1
S	BDCI2020 ^{EOD}	2020	1.3	RGB	145,981	256	2m	7
S	BHPools&WaterTanks ^{EOD}	2020	1.9	Optical (Aerial)	350	3000	–	1
S	DFC20 ^{EOD}	2020	19.2	Sentinel-1,Sentinel-2	180662	–	10m	17
S	DubaiSemSeg ^{EOD}	2020	0.032	Optical (Satellite)	72	800	–	6
S	EORSSD ^{EOD}	2020	0.06	RGB	2,000	500	–	2
S	FUSAR-Map ^{EOD}	2020	–	SAR (Satellite)	610	1024	–	4

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Table 1 continued

Task	Dataset	Year	GB	Modality / sensor	Samples	Sample size	Resolution	Classes
S	GTA-V SID ^{EOD}	2020	0.1	Optical (Satellite)	121	500	–	–
S	Ghana Crop Type ^{EOD}	2020	312	Sentinel-1, Sentinel-2, Planet	2881*14*25	64	3~10m	5
S	HKH Glacier Mapping ^{EOD}	2020	29.4	Multispectral (Satellite)	14190*15	512	–	2
S	Kag. Massachusetts Buildings ^{EOD}	2020	3	RGB	151	1500	1m	1
S	Kag. Satellite Buildings ^{EOD}	2020	0.878	Optical (Satellite)	6038	256	–	1
S	Kag. Satellite Water Bodies ^{EOD}	2020	0.274	Multispectral (Satellite)	2841	1000	–	1
S	Kaggle aerial segmentation ^{EOD}	2020	0.033	RGB	72	800	–	6
S	Kaggle buildings segmentation ^{EOD}	2020	0.899	RGB	6038	256	–	2
S	Kenya Crop Type	2020	–	Sentinel-2 MSI	4688	3035x2016	10m	7
S	LandCoverAI ^{EOD}	2020	1.4	RGB	41	9000	0.25m, 0.5m	–
S	Open Cities AI Challenge ^{EOD}	2020	81.5	RGB	11,000	1,024	0.03~0.2m	2
S	OpenSARUrban ^{EOD}	2020	2.2	SAR (Satellite)	33358	100	–	10
S	Parking AP-KLOT ^{EOD}	2020	3	RGB	501	–	–	2
S	Power TTPL ^{EOD}	2020	4.2	RGB	1100	3840	UAV	3
S	Rwanda Built-Up ^{EOD}	2020	0.07	Optical (Satellite)	787	256	–	1
S	SEN-12-FLOOD ^{EOD}	2020	12.2	RGB, SAR, MS	336	–	10m	1
S	SemCity Toulouse	2020	5.2	MS	16	3500	0.5~2m	–
S	South Sudan Crop Type ^{EOD}	2020	977	Multiple (Satellite)	687*14*25	64	–	5
S	SpaceNet-6	2020	0.368	SAR, RGB	3401	900	0.5m	1
S	SpaceNet-7	2020	20.6	MS, MT	1525	1024	4m	1
S	Synthinel-1 ^{EOD}	2020	1	RGB	2108	572	0.3m	1
S	TTPLA ^{EOD}	2020	4.2	Optical (Aerial)	1100	3840	–	–
S	TimeSen2Crop	2020	1.1	Sentinel-2	1000000	10980	10m	16
S	UAVid ^{EOD}	2020	5.88	RGB	420	4000	UAV	–
S	WHU-Hi ^{EOD}	2020	0.817	Hyperspectral (Drone)	3	1217	0.109m	11
S	WHU-Hi-HongHu ^{EOD}	2020	–	Hyperspectral	1	940x475	0.043m	22
S	Water Body Kaggle Water Bodies ^{EOD}	2020	0.28	RGB	2841	1000	–	2
S	iSAID	2020	6.54	RGB	2806	4000	–	15
S	Arctic Arctic Sea Ice Image Masking ^{EOD}	2021	0.092	RG-NIR	3392	357x306	10m	8
S	Benin Cashews	2021	–	Sentinel-2 (labels corrected with Pléiades)	70	1186x1122	10m imagery / 0.5m reference	6
S	DALES Object ^{EOD}	2021	5	ALS	492000000	–	50 pts/m2	8
S	DFC21-DSE ^{EOD}	2021	18	SAR, MS, Hyperspectral	6000	16	10~750m	4
S	DoriaNet ^{EOD}	2021	0.067	Optical (Drone)	2409	1280	–	3
S	ETCI2021 ^{EOD}	2021	5.3	SAR (Satellite)	66810	256	–	2
S	EuroCrops ^{EOD}	2021	8.6	Sentinel-2	805,401	0.5 ha	10m	43
S	FloatingObjects ^{EOD}	2021	15.4	Multispectral (Satellite)	–	–	–	–
S	FloodNet ^{EOD}	2021	12.1	RGB	2343	4000	0.015m	10
S	Hessigheim 3D ^{EOD}	2021	5.95	Drone	73909354	–	800 pts/m2	11
S	LoveDA	2021	9.6	RGB	5987	1024	0.3m	7
S	MARIDA	2021	1.2	Multispectral (Satellite)	1381*10	256	–	15
S	MUESLI ^{EOD}	2021	37.7	Multiple (Aerial)	1*415	12800	–	41

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Table 1 continued

Task	Dataset	Year	GB	Modality / sensor	Samples	Sample size	Resolution	Classes
S	Mini Inria Aerial Image Labeling Dataset ^{EOD}	2021	–	RGB	32,500	512	0.3m	2
S	MultimodalBen ^{EOD}	2021	0.489	Multiple (Multiple)	568	680	–	30
S	PASTIS	2021	29	Sentinel-2	2433*45*10	128	10m	18
S	PASTIS-R	2021	54	Multiple (Satellite)	2433	128	–	18
S	RescueNet ^{EOD}	2021	21.8	Optical (Drone)	4494	3500	–	12
S	S2-SHIPS ^{EOD}	2021	2.9	Multispectral (Satellite)	16*16	1783	–	3
S	Sen4AgriNet ^{EOD}	2021	262	Sentinel-2	–	–	10~60m	–
S	SmallholderPlant ^{EOD}	2021	1	Multispectral (Satellite)	70*13	1180	–	6
S	The Azavea Cloud Dataset ^{EOD}	2021	–	Sentinel-2	32	–	10m~60m	2
S	UOPNOA ^{EOD}	2021	9.7	Multiple (Multiple)	36000	–	–	–
S	AIR-PolSAR-Seg ^{EOD}	2022	0.609	SAR	2000	512	8m	6
S	Burned Area Delineation ^{EOD}	2022	14	Multiple (Satellite)	–	–	–	–
S	C2S-MS ^{EOD}	2022	12	Multiple (Satellite)	1800	512	–	3
S	DFC22-SSL ^{EOD}	2022	42	Multiple (Aerial)	5000	2000	–	14
S	FLAIR #1 ^{EOD}	2022	70	Optical (Aerial)	77412	512	–	13
S	FireCube ^{EOD}	2022	23.5	Multiple (Multiple)	4314*58	700	–	1
S	Five-Billion-Pixels ^{EOD}	2022	104	RGB,MS	150	7200×6800	4m	24
S	GeoNRW ^{EOD}	2022	30	RGB,nDSM	7783*3	1000	1m	10
S	GrowliFlower ^{EOD}	2022	11.7	Multispectral (Drone)	8402	256	–	4
S	HOSD ^{EOD}	2022	2.2	Hyperspectral (Aerial)	18*224	1700	–	1
S	HelixNet ^{EOD}	2022	235.7	MLS	8850000000	–	–	9
S	Landslide4Sense ^{EOD}	2022	2.4	Multiple (Satellite)	4844*14	128	–	1
S	MMFlood ^{EOD}	2022	11.3	SAR (Satellite)	1748*2	2000	–	1
S	MapInWild ^{EOD}	2022	365	Multiple (Satellite)	1018*8	1920	–	11
S	Microsoft RoadDetections ^{EOD}	2022	9.25	RGB	20000	1088	1m	1
S	MiniFrance ^{EOD}	2022	93.1	RGB	2121	10000	0.5m	12
S	MultiSenGE ^{EOD}	2022	530	Sentinel-1, Sentinel-2	1084260*12	256	10m	14
S	NASA-Harvest ^{EOD}	2022	0.3	Multispectral (Satellite)	70	256	–	1
S	NDWS ^{EOD}	2022	2	Multiple (Multiple)	18445*12	64	–	1
S	OnCloudN ^{EOD}	2022	50	Optical (Satellite)	22728	512	–	1
S	OpenEarthMap	2022	9.1	Optical (Multiple)	5000	1024	–	8
S	OpenSentinelMap ^{EOD}	2022	445	RGB, Sentinel-2	137045	192	10m~60m	15
S	RDCME ^{EOD}	2022	–	Optical (Satellite)	775	256	–	1
S	RID ^{EOD}	2022	1.5	RGB	2000	–	0.1m	16
S	S12 ScotInd Cloud ^{EOD}	2022	0.96	Multiple (Satellite)	–	–	–	1
S	STPLS3D ^{EOD}	2022	36.6	Photogrammetry	–	–	–	20
S	Sat Burn Area ^{EOD}	2022	14	Multispectral (Satellite)	73*14	1500	–	1
S	SeasoNet	2022	229	Sentinel-2	1759830	120	10m	33

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Table 1 continued

Task	Dataset	Year	GB	Modality / sensor	Samples	Sample size	Resolution	Classes
S	Sentinel-2 Cloud Cover ^{EOD}	2022	51.2	MS	22728	–	10m~60m	2
S	SpaceNet-8 ^{EOD}	2022	6.8	Panchromatic,RGB	1200*2	1200	0.3~0.8m	2
S	TimeSpec4LULC ^{EOD}	2022	60	MS,MT	–	262 months	500m	29
S	UBC ^{EOD}	2022	0.675	Optical (Satellite)	800	600	–	61
S	WHU-OHS ^{EOD}	2022	94.9	Hyperspectral	7795	512	10m	24
S	WildForest3D ^{EOD}	2022	0.081	Drone	7000000	–	60 pts/m2	6
S	WildfireDanger ^{EOD}	2022	17	Multiple (Multiple)	–	–	–	1
S	GAMUS ^{EOD}	2023	48	RGB,nDSM	9340	1024	0.33m	6
S	LuoJia-HSSR ^{EOD}	2023	–	RGB	6438	256	0.75m	23
S+C	DeepGlobe-Bldng ^{EOD}	2018	–	Optical (Satellite)	24586	650	–	1
S+C	SEN12MS	2019	510	MS,SAR	180662	256	10m	–
S+C	FLAME ^{EOD}	2020	43.7	Multispectral (Drone)	47992	3840	–	2
S+C	S2 Cloud Mask Catalogue ^{EOD}	2020	15.4	Sentinel-2	513*13	1022	20m	25
S+C	DFC23 ^{EOD}	2023	–	Multiple (Satellite)	–	–	–	12
SOIL	Hyperview Challenge ^{EOD}	2022	1.85	Hyperspectral	2886	11-250	2m	–
SR	SEN2VENmuS	2022	87	Sentinel-1, Sentinel-2	132955	256	10-20m	–
SR	The WorldStrat	2022	109.5	MS, Sentinel-2, RGB	3449	1054	1.5-60m	8
SR/PSDFC06 ^{EOD}		2006	0.39	Multispectral (Aerial)	6	5000	–	–
SR/PSProba-V Super Resolution		2018	0.692	Multispectral (Satellite)	1160	384	–	–
SR/PSALSAT-2B		2021	0.07	Optical (Satellite)	5518	256	–	–
SR/PSPAirMAX (Airbus) ^{EOD}		2021	0.153	Multispectral (Satellite)	14	–	–	–
SR/PSPushFrameSR ^{EOD}		2021	1.1	Optical (Satellite)	960*15	256	–	–
SR/PSJilin1 VidSR ^{EOD}		2022	19.4	Optical (Satellite)	6250	3000	–	–
SR/PSMu-S2 ^{EOD}		2022	2.8	Multispectral (Satellite)	1274	100	–	–
SR/PSMultiExposureSR ^{EOD}		2022	0.743	Optical (Satellite)	–	–	–	–
SR/PSPRISMA Pan Sharp ^{EOD}		2022	1.7	Hyperspectral (Satellite)	4	2400	–	–
SR/PSSSEN2VENμS		2022	85	Multispectral (Satellite)	132955	256	–	–
SR/PSWorldStrat ^{EOD}		2022	107	Optical (Satellite)	3928	1054	–	–
3DPC	Oakland 3-D PointCloud ^{EOD}	2009	0.033	PointCloud	1.6M	1.5 km	–	5
3DPC	Paris-rue-Madame ^{EOD}	2014	–	PointCloud	20M	1 km	–	26
3DPC	District of Columbia LiDAR 2015 ^{EOD}	2015	34.4	PointCloud	–	–	–	8
3DPC	Paris-Lille-3D ^{EOD}	2017	19.9	PointCloud	143M	1.94 km2	–	50
3DPC	Semantic3D.net ^{EOD}	2017	24.5	PointCloud	4000M	–	–	8
3DPC	CORE3D ^{EOD}	2018	–	Panchromatic, MS, PointCloud	–	–	–	–
3DPC	District of Columbia LiDAR 2018 ^{EOD}	2018	279.8	PointCloud	–	–	–	11
3DPC	DublinCity ^{EOD}	2019	166.8	PointCloud	260M	2.0 km2	–	13
3DPC	SemanticKITTI ^{EOD}	2019	80.2	PointCloud	4549M	39.2 km2	–	25
3DPC	Campus3D ^{EOD}	2020	–	PointCloud	937.1M	1.58 km2	UAV	14
3DPC	DALES ^{EOD}	2020	–	PointCloud	505.3M	10.0 km2	–	8

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Table 1 continued

Task	Dataset	Year	GB	Modality / sensor	Samples	Sample size	Resolution	Classes
3DPC	LASDU ^{EOD}	2020	–	PointCloud	3.12M	1.02 km2	Aircraft @ 1200m	5
3DPC	SensatUrban ^{EOD}	2020	20.6	PointCloud	2847.1M	6 km2	UAV	13
3DPC	Swiss3DCities ^{EOD}	2020	–	PointCloud	226M	2.7 km2	–	5
3DPC	Toronto3D ^{EOD}	2020	1.1	PointCloud	78.3M	1.0 km2	–	8
3DPC	Heisenheim3D ^{EOD}	2021	58.8	PointCloud	125.7M	0.19 km2	–	11
3DPC	SUM-Helsinki ^{EOD}	2021	8.35	Mesh	19M	4 km2	0.075m	6
3DPC	benchmark_ISPRS2021 ^{EOD}	2021	3.22	RGB	20	1024	0.08m	–
VQA	RSIVQA ^{EOD}	2021	29.5	RGB	37264	256-4000	0.1-8m	864
VQA+	CDVQA ^{EOD}	2022	–	RGB	2968	512	0.5-3m	19
VEH	AM3D-Real ^{EOD}	2022	–	RGB, PointCloud	1012	720x480	0.25m	2
WB	Forel-Ule Index global inland waters ^{EOD}	2021	0.004	MS	–	–	500m	–
WTHR	Historical Hourly Weather ^{EOD}	2017	0.075	Weather attributes	–	–	–	–
WTHR	ClimateNet ^{EOD}	2020	28	NC variables	459	16 (768,1152)	–	9
WTHR	DeepWeather ^{EOD}	2020	2,965	Weather attributes	20	–	–	9

Task codes: C = classification; S = semantic segmentation; OD = object detection; CD = change detection; R = regression; SR = super-resolution; PS = pan-sharpening; SSL = self-supervised pretraining; CR = cloud removal; M3D = multiview 3D; 3DPC = urban 3D point cloud; IC = image captioning; IM = image matching; IR = image registration; IT = image translation; GL = geo-localization; VQA = visual question answering. Combined codes (e.g. CD+C, OD+S, S+C) indicate multiple tasks. Less common domain-specific codes: AE = action event; AG = agriculture; AQ = air quality; AD = anomaly detection; BLD = building; CHR = chronology; CLIM = climate; DH = dehazing; FLT = filtering; FOR = forest; GEO = geophysical; O = other; PT = plant/tree; POP = population; POV = poverty; SOIL = soil parameter; VEH = vehicles; WB = water body; WTHR = weather.

2.3 Dataset and Benchmark Suites

Suite	Year	Type	Scale	Tasks	Modalities
EarthNets bench-mark	2022	Evaluation bench-mark / platform	5 selected large-scale datasets	Classification, detection, segmentation, change detection	Multiple EO modalities
GEO-Bench	2023	GFM evaluation suite	12 subdatasets	6 classification + 6 semantic segmentation	RGB, SAR, multispectral, hyperspectral
SATIN	2023	Remote-sensing evaluation suite	27 datasets	Classification (6 task groups, >250 labels)	Primarily RGB
SustainBench	2021	Sustainable-development bench-mark suite	15 tasks	Regression, classification, time-series and related tasks	EO imagery auxiliary socioeconomic/environmental data

Table 2: Dataset and benchmark suites identified in the reviewed literature up to 2023. Suite names link to their project pages.

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